Report: Responsible and Effective Use of AI in Academic and Technical Work

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1. Introduction

Artificial Intelligence (AI), particularly Large Language Models (LLMs), has become an essential part of daily life and technical workflows. However, as highlighted in the concept of **Responsible AI** and the **CS10014 AI Usage Guidelines**, AI is designed to support and enhance human decision-making rather than replace human judgment or academic effort. Using AI ethically requires a strong commitment to transparency, accountability, academic integrity, and a deep understanding of the final work.

2. Have You Used AI Before?

Yes, I have used AI tools (such as Gemini, ChatGPT, and GitHub Copilot) regularly in my studies and personal programming projects. Rather than using AI to bypass learning or auto-generate complete assignments, I integrate it as a learning assistant across four main areas aligned with the course's permissible use cases:

1. **Knowledge Lookup and Concept Clarification:**
   When studying complex algorithms, data structures, or mathematical concepts, I use AI to break down abstract ideas into simpler explanations or analogies before reading the formal proofs in textbooks.
2. **Code Debugging and Optimization Suggestions:**
   During programming tasks (such as in C++), I ask AI to suggest implementation structures, help identify subtle logic bugs, or analyze time and space complexity. Afterward, I write and validate the actual code using my own test cases.

3. **Generating Simulated Test Data:**
   I use AI to quickly generate small edge-case datasets (e.g., sample graphs, JSON/CSV inputs) to test the robustness of my deterministic algorithms.
4. **Report Structuring and Language Refinement:**
   When writing technical documentation in Markdown or LaTeX, I use AI to check English grammar, refine phrasing for academic professionalism, and suggest clean table or report layouts.

3. Experience and Tips for Using AI Effectively and Responsibly

Based on my practical experience and the principles from the provided documents (*Responsible AI* and *CS10014 AI Usage Guidelines*), here are my core tips for using AI effectively:

Tip 1: Apply "No Blind Copying" and Prioritize Deep Understanding

- **Experience:** AI-generated code or text can look convincing on the surface but fail on edge cases or contain logical flaws.
- **Actionable Tip:** Never copy and paste AI outputs directly into a project. Always rewrite explanations in your own words and ensure you can explain every line of code and algorithmic step during an in-class or oral examination.

Tip 2: Cross-Verify Everything (Academic Integrity & Bias Mitigation)

- **Experience:** Because AI models are trained on vast datasets, they can suffer from data bias or "hallucinate" inaccurate references and formulas.
- **Actionable Tip:** Treat AI as a starting point, not an official academic source. Always cross-verify AI explanations against authoritative textbooks, official documentation, and empirical benchmarks.

Tip 3: Use Structured, Step-by-Step Prompts (Explainability)

- **Experience:** Asking vague prompts like *"Solve this problem for me"* leads to poor learning outcomes and generic results.
- **Actionable Tip:** Decompose the problem first. Use structured prompts (such as Chain-of-Thought prompting) to ask AI *how* and *why* a specific method works (e.g., *"Compare the trade-offs*

between BFS and DFS for this constraint"). This improves explainability and keeps you in control of the system design.

Tip 4: Maintain Transparency and Accountability (Document Your AI Usage)

- **Experience:** As developers and students, we are 100% accountable for the code and reports we submit, regardless of whether AI assisted in the process.
- **Actionable Tip:** Keep a clear log of your AI interactions—including the tool name, version, access time, exact prompts, and how you modified or validated the output. Always declare this information honestly in the **Appendix (AI Usage Notes)** of your course reports.

Tip 5: Respect Data Privacy and Security

- **Experience:** Responsible AI requires strict protection of personal and sensitive data.
- **Actionable Tip:** Never input private personal information, credentials, or confidential data into public AI platforms.

4. Conclusion

AI is a powerful multiplier for productivity when used with discipline. By combining the ethical framework of **Responsible AI** (transparency, fairness, accountability, and privacy) with the strict academic boundaries of our **Course Guidelines**, we can leverage AI to sharpen our Computational Thinking skills without compromising our independent learning ability.